TRT 903 · DOCTORAL TEXTBOOK

Advanced Neurobiology of Trauma

How threat, adversity and recovery are written into the nervous system

Why do some experiences leave lasting marks on the brain and body while others do not? This doctoral-level textbook follows trauma through every level of the nervous system — from molecules, synapses and genes to circuits, stress physiology, memory, sleep and social context — and asks at each step what the evidence actually shows, what it merely suggests, and what it cannot support.

Written for critical readers, each chapter pairs concise, source-verified exposition with learning objectives, synthesis tables, composite teaching cases, review questions and research assignments. A 187-term glossary, laboratory and classroom teaching protocols, assessment checklists and blank learner worksheets support structured study and supervision.

INSIDE THIS BOOK

- Foundations, functional neuroanatomy and lifespan brain development
- Cellular and molecular neuroscience, neurotransmission and neuroplasticity
- HPA-axis stress physiology, allostasis, genetics and epigenetics
- Threat learning, traumatic memory, emotion regulation and dissociation
- Neuroimmunology, neuroimaging methods and their limits
- Sleep and circadian rhythms, developmental and structural trauma
- Assessment, differential diagnosis and therapeutic translation
- Research methods, evidence appraisal and professional integration

For doctoral students, clinicians, researchers and educators working at the interface of neuroscience and traumatic stress.

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